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Regression Splines

Basis Functions

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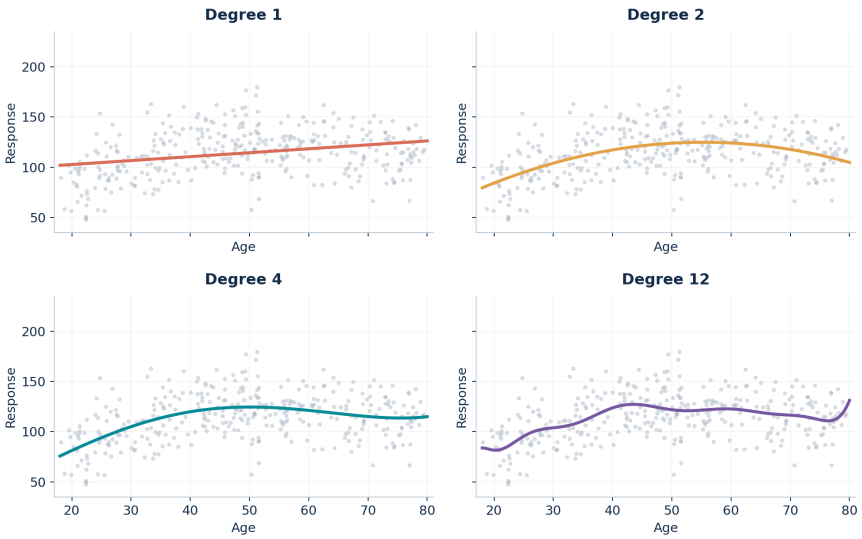
- Polynomial and piecewise-constant regression models are in fact special cases of a **basis function approach**.

$$f(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x) + \cdots + \beta_K b_K(x)$$

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Polynomial Regression

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Basis Functions

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- Polynomial and piecewise-constant regression models are in fact special cases of a **basis function approach**.

$$f(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x) + \cdots + \beta_K b_K(x)$$

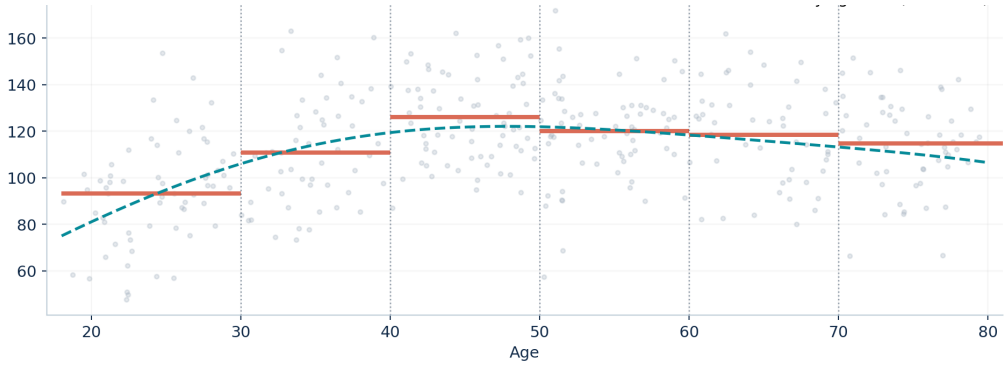
$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots + \beta_d x^d$$

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Step Functions

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$$f(x) = \beta_0 + \beta_1 C_1(x) + \beta_2 C_2(x) + \cdots + \beta_K C_K(x)$$



Basis Functions

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- Polynomial and piecewise-constant regression models are in fact special cases of a **basis function approach**.

$$f(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x) + \cdots + \beta_K b_K(x)$$

$$f(x) = \beta_0 + \beta_1 C_1(x) + \beta_2 C_2(x) + \cdots + \beta_K C_K(x)$$

Piecewise Polynomials

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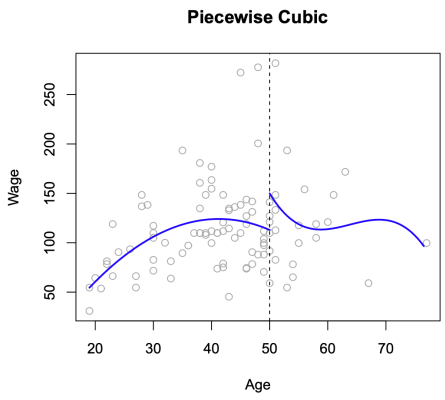
- Instead of fitting a high-degree polynomial over the entire range of X, piecewise polynomial regression involves fitting separate low-degree polynomials over different regions of X.
- A piecewise cubic polynomial with a single knot at a point c:

$$f(x) = \begin{cases} \beta_{01} + \beta_{11}x + \beta_{21}x^2 + \beta_{31}x^3 & \text{if } x < c \\ \beta_{02} + \beta_{12}x + \beta_{22}x^2 + \beta_{32}x^3 & \text{if } x \geq c \end{cases}$$

Piecewise Polynomials

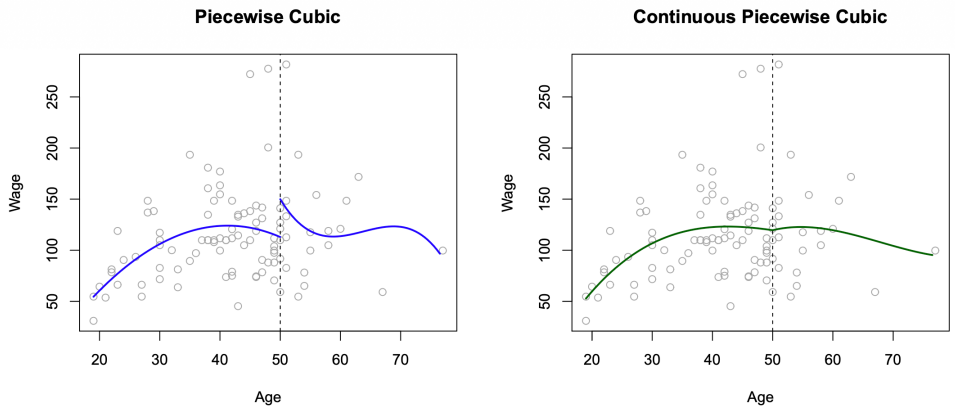
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$$f(x) = \begin{cases} \beta_{01} + \beta_{11}x + \beta_{21}x^2 + \beta_{31}x^3 & \text{if } x < c \\ \beta_{02} + \beta_{12}x + \beta_{22}x^2 + \beta_{32}x^3 & \text{if } x \geq c \end{cases}$$



Piecewise Polynomials

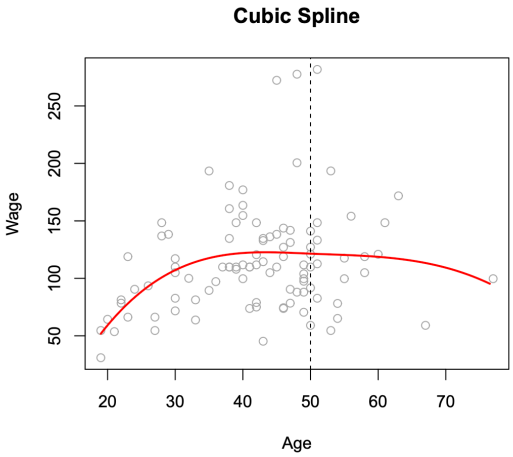
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Piecewise Polynomials

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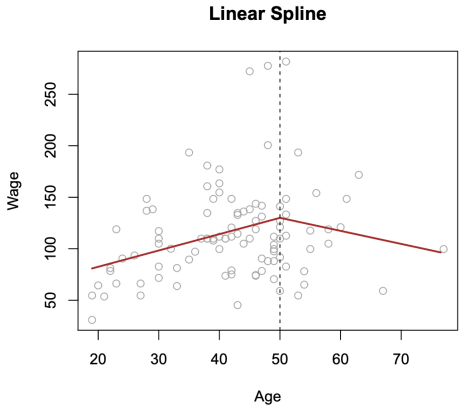
First derivative and second order derivative need to be constrained



Piecewise Polynomials

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d-1 continuity?
zero continuity - just continuity... not differntiation



Linear Splines

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- A linear spline with knots at ξ_k, k = 1, . . . , K, is a piecewise linear polynomial continuous at each knot.

$$f(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x) + \cdots + \beta_K b_K(x)$$

$$\begin{aligned} b_1(x) &= x \\ b_{k+1}(x) &= (x - \xi_k)_+, \quad k = 1, \ldots, K \end{aligned}$$

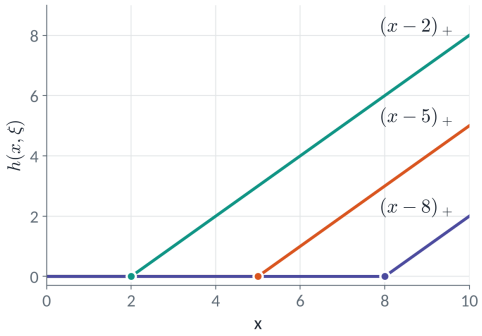
hinge function
when x > knot(xi), use that function
otherwise, 0

Linear Splines

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$$h(x,\xi) = (x-\xi)_+ = \max(0,\ x-\xi) = \begin{cases} x-\xi & \text{if } x > \xi \\ 0 & \text{otherwise} \end{cases}$$

- **Zero to the left of the knot:** it has no effect there.
- **Slope = 1 to the right:** it adds a straight ramp.
- **Continuous everywhere:** no jump at ξ, only a bend (kink).
- Also called a **truncated power basis** function of degree 1 (ReLU in deep learning).

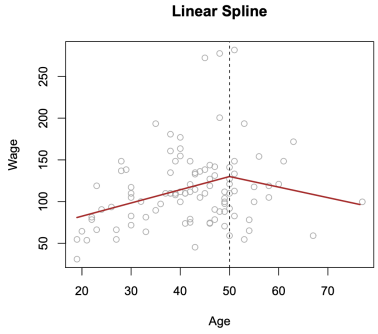


Linear Splines

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- A linear spline with knots at ξ_k, k = 1, . . . , K, is a piecewise linear polynomial continuous at each knot.

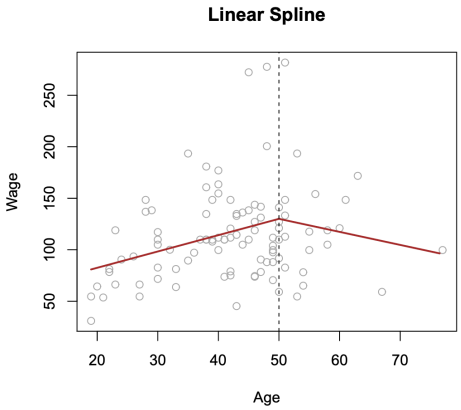
$$f(x) = \beta_0 + \beta_1 x + \beta_2 (x-\xi_1)_+ + \cdots + \beta_{K+1}(x-\xi_K)_+$$



Linear Splines

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f(x) = b0 + b1x where x < knot(xi)
f(x) = b0 + b1x + b2(x-knot) where x > knot(xi)



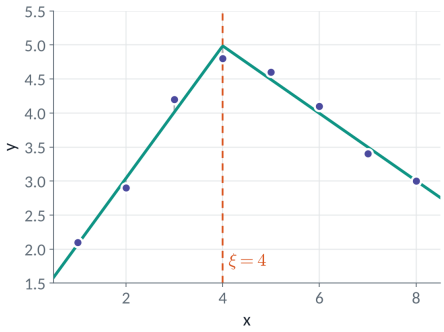
Intercepts and slopes are different!!

Linear Splines

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$$f(x) = \beta_0 + \beta_1 x + \beta_2 (x-4)_+$$

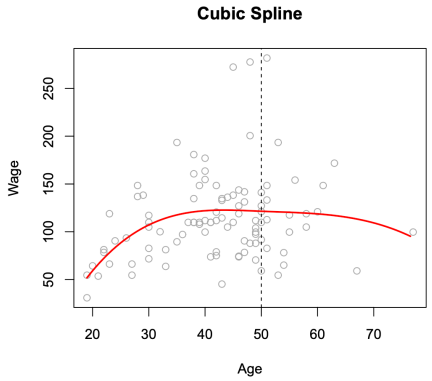
x	y	$(x-4)_+$
1	2.1	0
2	2.9	0
3	4.2	0
4	4.8	0
5	4.6	1
6	4.1	2
7	3.4	3
8	3.0	4



Cubic Splines

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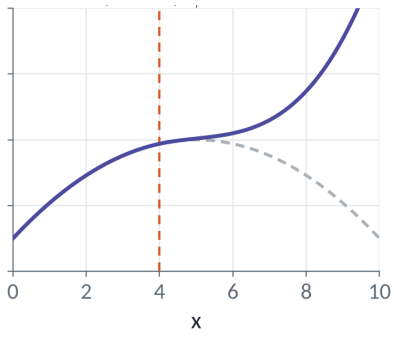
$$f(x) = \begin{cases} \beta_{01} + \beta_{11}x + \beta_{21}x^2 + \beta_{31}x^3 & \text{if } x < c \\ \beta_{02} + \beta_{12}x + \beta_{22}x^2 + \beta_{32}x^3 & \text{if } x \geq c \end{cases}$$



Cubic Splines

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$$f(x) = \beta_0 + \beta_1 b_1(x) + \beta_2 b_2(x) + \beta_3 b_3(x) + \cdots + \beta_K b_K(x)$$



Cubic Splines

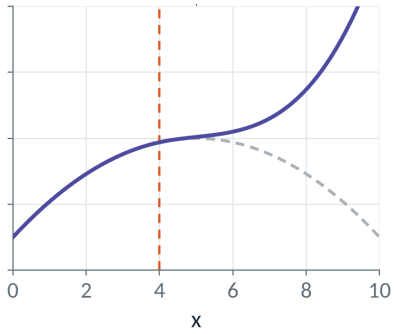
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Multiple linear regression!

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - c)_+^3$$

Why (x-c)^3? To satisfy the continuity of first- and second-order derivative

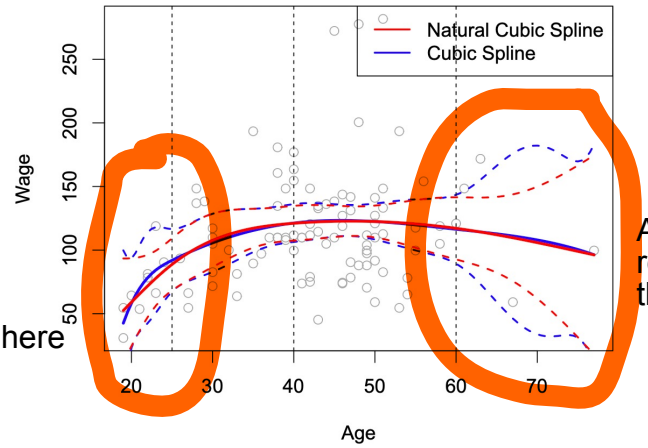
x	y	x^2	x^3	$(x-4)^3_+$
1	2.1			0
2	2.9			0
3	4.2			0
4	4.8			0
5	4.6			1
6	4.1			8
7	3.4			27
8	3.0			64



Natural Cubic Splines

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Much more frequently used!



2 parameters here

And 2 parameters here removed when we fit the natural spline

If there is a small sample in the extreme,
just let them 'linear'. -> give this constraint in the boundary location!

e.g., age < 20 and age > 60

Simpler function can reduce the variance.

Choosing the Number and Locations of the Knots

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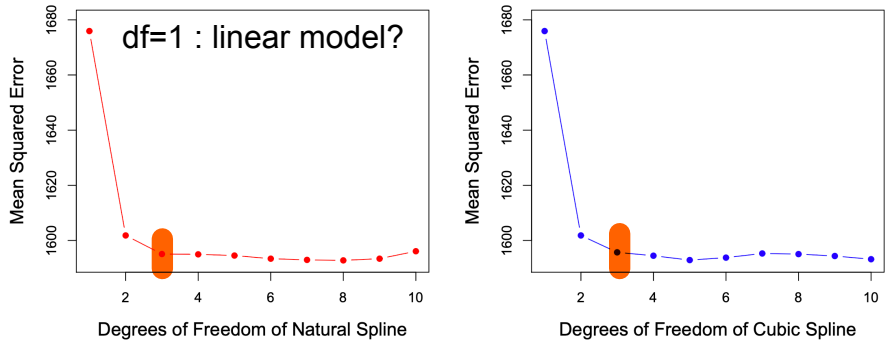
Is this equal to K?

- One strategy is to decide ***K***, the **number of knots**, and then place them at appropriate quantiles of the observed X.
- A cubic spline with *K* knots has *K* + 4 parameters (df).
- A natural spline with *K* knots has *K* parameters (df).

I think it is quite hard to understand the concept of 'df' here...
It depends on the software package and the mathematical definition;

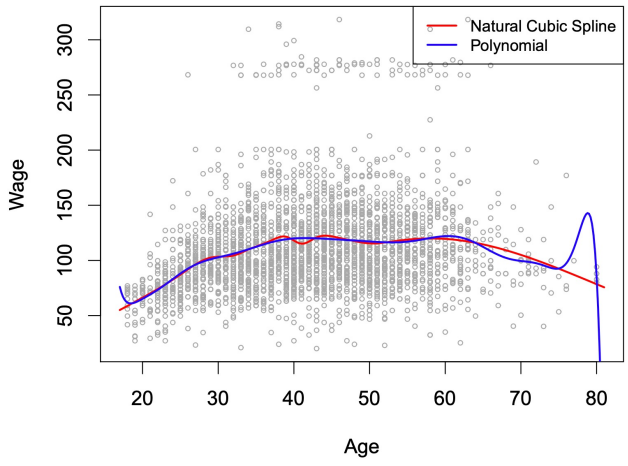
Choosing the Number and Locations of the Knots

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Choosing the Number and Locations of the Knots

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left and right extreme boudary are different between natural cubic and polynomial

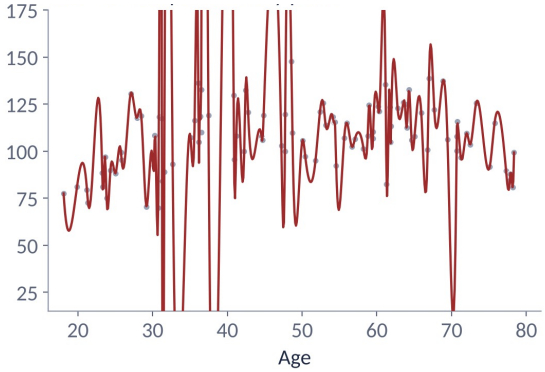
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Smoothing Splines

Smoothing Splines

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$$\text{minimize} \quad \sum_{i=1}^{n}(y_i - g(x_i))^2$$



Smoothing Splines Minimize this 'regularized' target

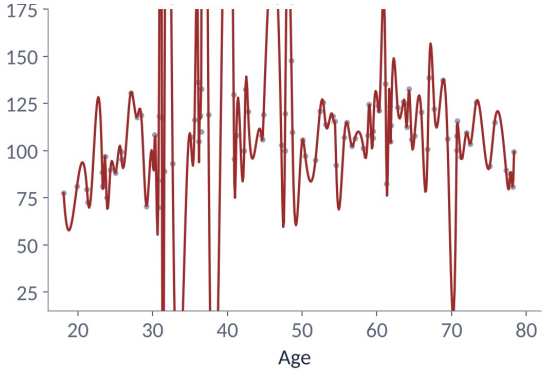
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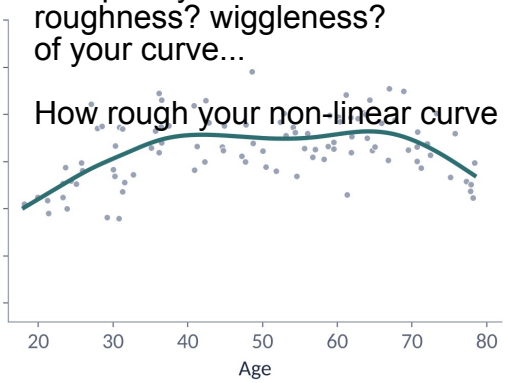
small prediction error Penalty term

$$\text{minimize} \quad \sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

Complexity?
roughness? wiggleness?
of your curve...

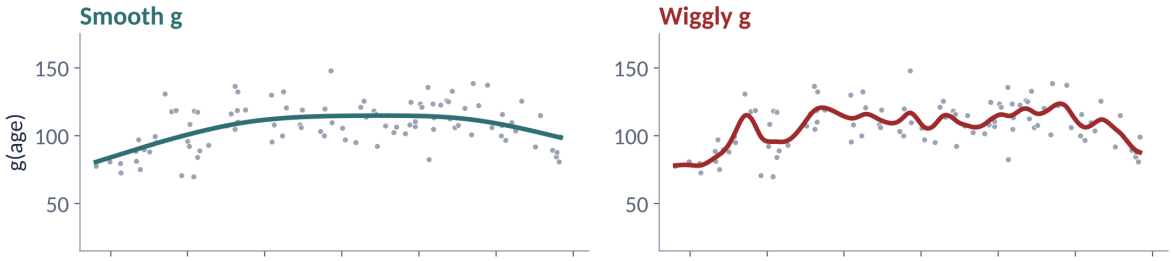
How rough your non-linear curve is.





Smoothing Splines

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$$\int g''(t)^2 dt = 0.23$$

Ong

$$\int g''(t)^2 dt = 1253$$

Smoothing Splines

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- Regression splines control flexibility through the knots. Smoothing splines control it through a penalty.

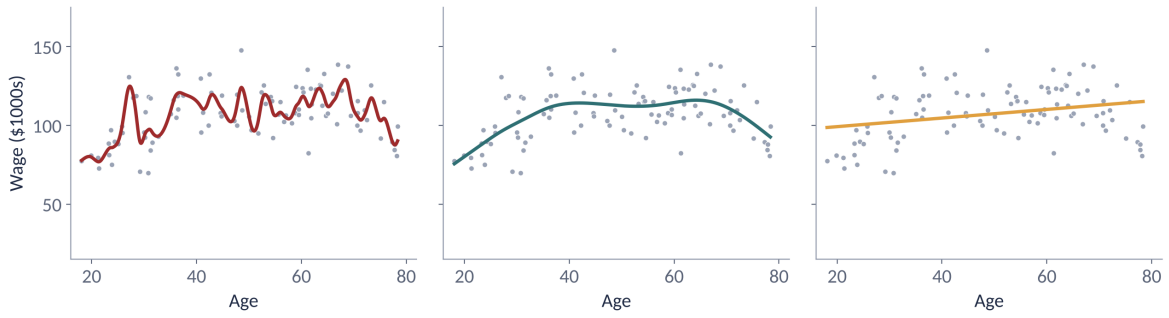
$$\text{minimize} \quad \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

- The smaller λ, the more wiggly the function, eventually interpolating y_i when λ = 0.
- As λ→∞, the function g(x) becomes linear.

Smoothing Splines

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small lambda large lambda



Smoothing Splines

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For every x values become the knot!!! (Someone mathematically proved)

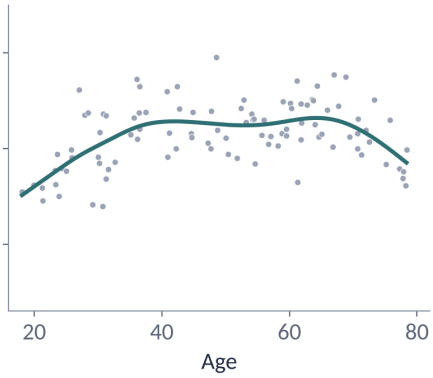
- The function $g(x)$ that minimizes the following is actually a **natural cubic spline** with knots at every unique value of x_i.
- It is not the same natural cubic spline obtained via the basis function approach; it is a shrunken version of a natural cubic spline, where the value of the tuning parameter λ controls the level of shrinkage.
- Smoothing splines avoid the knot-selection issue, leaving a single λ to be chosen.

$$\text{minimize} \quad \sum_{i=1}^{n}(y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

Smoothing Splines

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- The function $g(x)$ that minimizes the following is actually a **natural cubic spline** with knots at every unique value of x_i.



Smoothing Splines

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effective degree of freedom: larger lambda, degree of freedom will decrease (became more simpler)
small lambda, degree of freedom will increase (i.e., become complex)

- Although a smoothing spline has n parameters, these n parameters are heavily constrained. Hence df_λ is a measure of the flexibility of the smoothing spline—the higher it is, the more flexible (and the lower-bias but higher-variance) the smoothing spline.

S matrix only depends on X variable (Not Y)
S matrix is similar to Hat matrix in multiple regression

$$\hat{\mathbf{Y}} = \hat{\mathrm{g}}(\mathbf{X})_\lambda = \mathbf{S}_\lambda(\mathbf{X})\mathbf{Y}$$

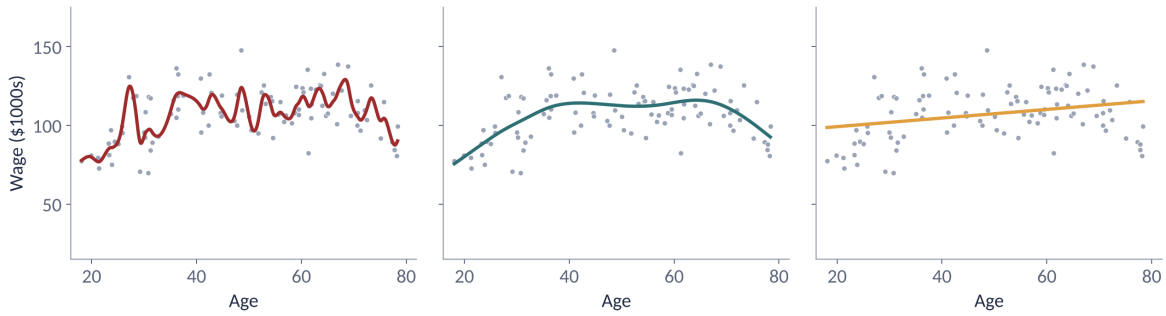
$$df_\lambda = \sum_{i=1}^{n} \{\mathbf{S}_\lambda\}_{ii}$$

effective degree of freedom = trace of the S matrix!

Smoothing Splines

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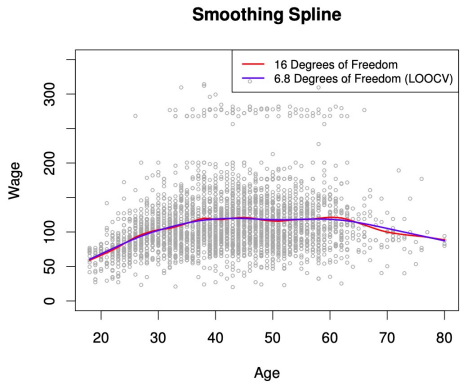
df = big!!! df = small



Smoothing Splines

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- We can find the value of λ that makes the cross-validated RSS as small as possible.
- It turns out that the LOOCV error can be computed very efficiently for smoothing splines, with essentially the same cost as computing a single fit.



We could have a non-interger degree of freedom here:
Why? this is the trace of the S matrix: and the diagonal element could be non-integer.

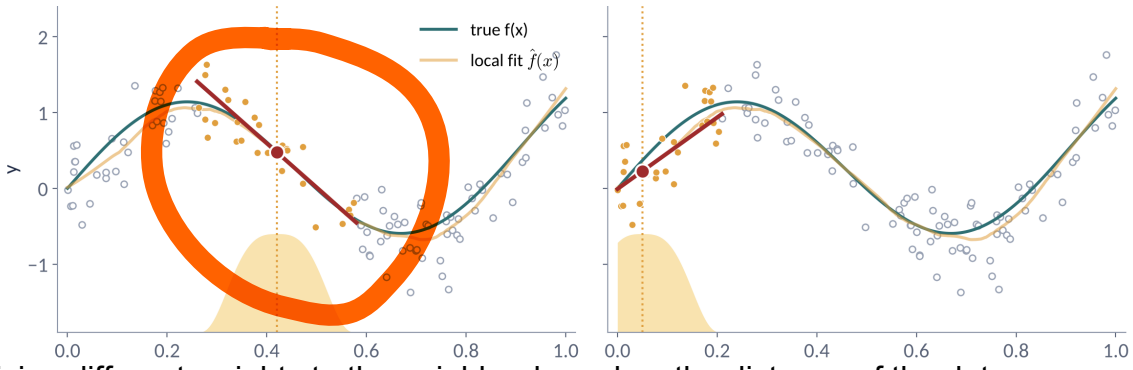
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Local Regression

Local Regression

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- Local regression involves computing the fit at a target point x0 using only the nearby training observations.



giving different weights to the neighbor based on the distance of the data.

Local Regression

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Algorithm 7.1 *Local Regression At $X = x_0$*

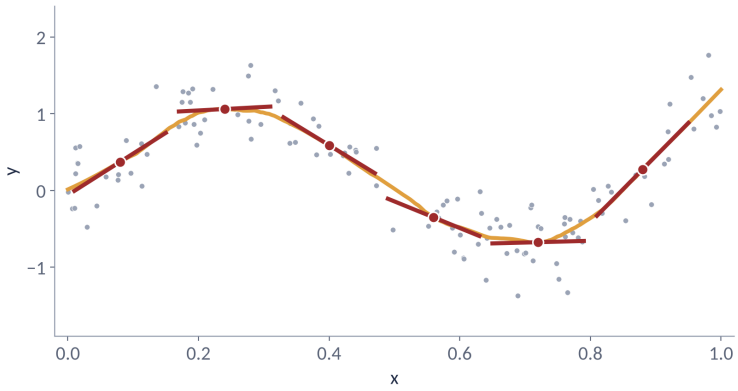
1. Gather the fraction $s = k/n$ of training points whose x_i are closest to x_0.
2. Assign a weight $K_{i0} = K(x_i, x_0)$ to each point in this neighborhood, so that the point furthest from x_0 has weight zero, and the closest has the highest weight. All but these k nearest neighbors get weight zero.
3. Fit a *weighted least squares regression* of the y_i on the x_i using the aforementioned weights, by finding $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize

$$\sum_{i=1}^{n} K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2. \tag{7.14}$$

4. The fitted value at x_0 is given by $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Local Regression

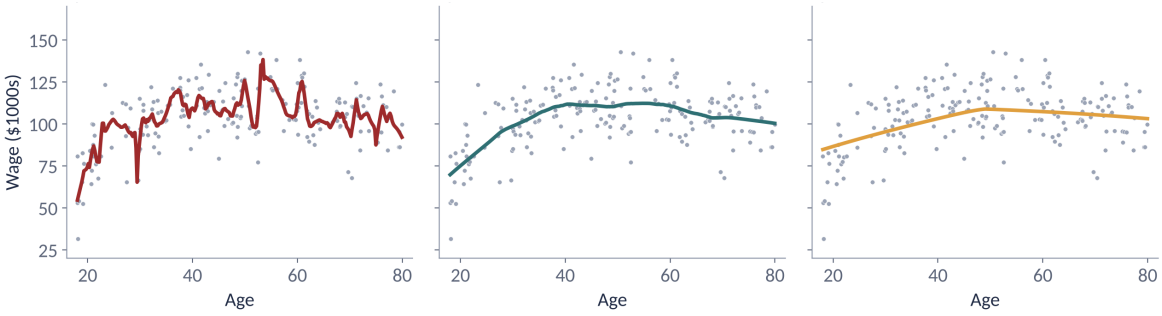
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Local Regression

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- The span *s* controls model flexibility.



smaller k, smaller s
more localize, more flexible

larger k, larger span s
more smooth.

Local Regression

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- Theoretically the same approach can be implemented in higher dimensions, using linear regressions fit to p-dimensional neighborhoods.
- However, local regression can perform poorly if p is much larger than about 3 or 4 because there will generally be very few training observations close to x0.
- KNN regression suffers from a similar problem in high dimensions.

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Generalized Additive Models

Generalized Additive Models

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- Generalized additive models (GAMs) provide a general framework for extending a standard linear model by allowing non-linear functions of each of the variables, while maintaining additivity.
- It replace each linear component in linear regression with a (smooth) non-linear function.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

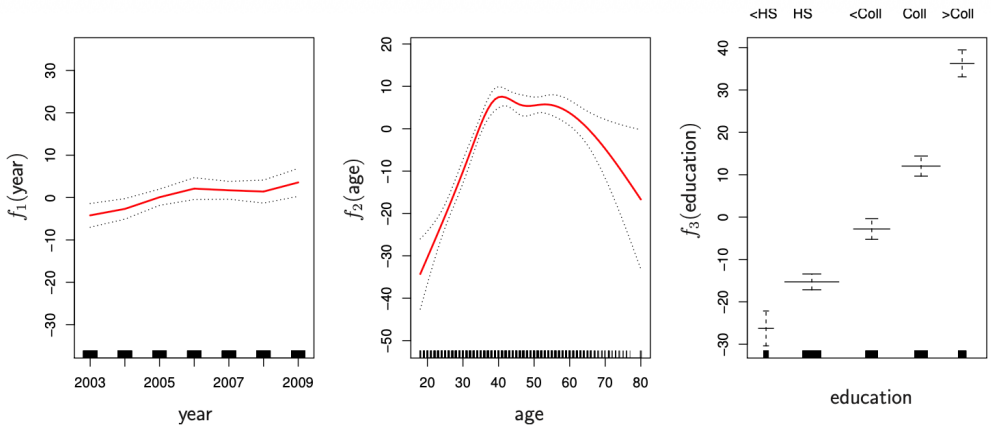
Generalized Additive Models

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- Any one-variable smoother can be a building block:
  - Step functions
  - Polynomials
  - Regression splines
  - Natural splines
  - Smoothing splines
  - Local regression

Generalized Additive Models

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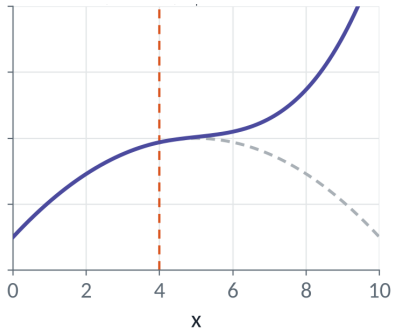
$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

Generalized Additive Models

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$$f_1(x_1)=b_0 + b_1x_1 + b_2{x_1}^2 + b_3{x_1}^3 + b_4(x_1 - c)^3_+$$

x	y	x^2	x^3	$(x-4)^3_+$
1	2.1			0
2	2.9			0
3	4.2			0
4	4.8			0
5	4.6			1
6	4.1			8
7	3.4			27
8	3.0			64



Generalized Additive Models

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$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$



$$f_1(\text{year}) = \sum_{k=1}^{4} \beta_{1k} B_{1k}(\text{year})$$

Generalized Additive Models

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$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

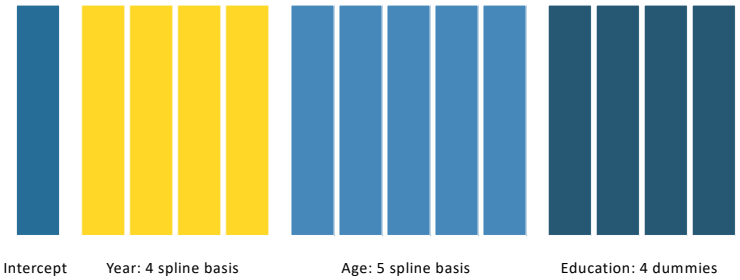


$$f_2(\text{age}) = \sum_{k=1}^{5} \beta_{2k} B_{2k}(\text{age})$$

Generalized Additive Models

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$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

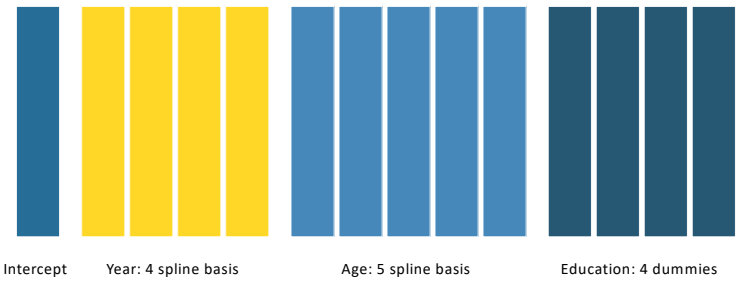


$$f_3(\text{education}) = \sum_{k=2}^{5} \gamma_k D_k$$

Generalized Additive Models

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$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$



$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

X

GAM: Smoothing Splines

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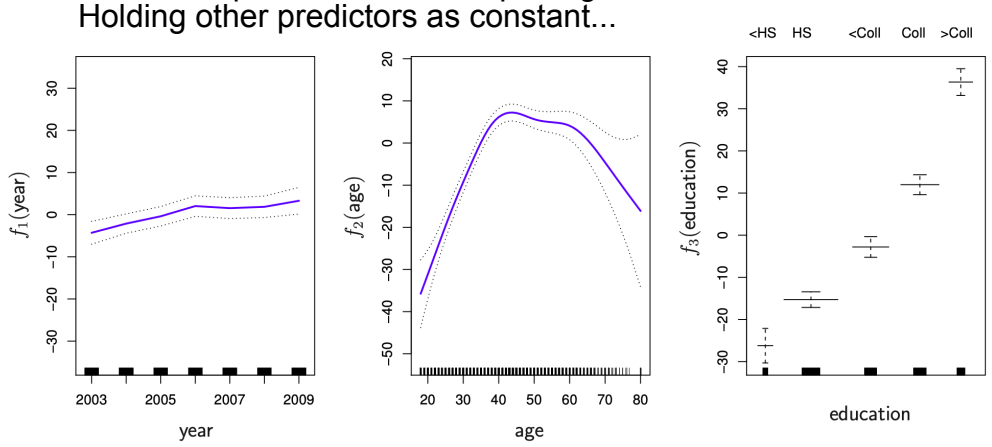
minimize

$$\sum_{i=1}^{n} \left[y_i - \beta_0 - f_1(t_i) - f_2(a_i) - f_3(e_i) \right]^2 + \lambda_1 \int [f_1''(t)]^2 dt + \lambda_2 \int [f_2''(a)]^2 da.$$

GAM: Smoothing Splines

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Same interpretation with multiple regression:
Holding other predictors as constant...



$$\hat{y} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

GAM: Classification

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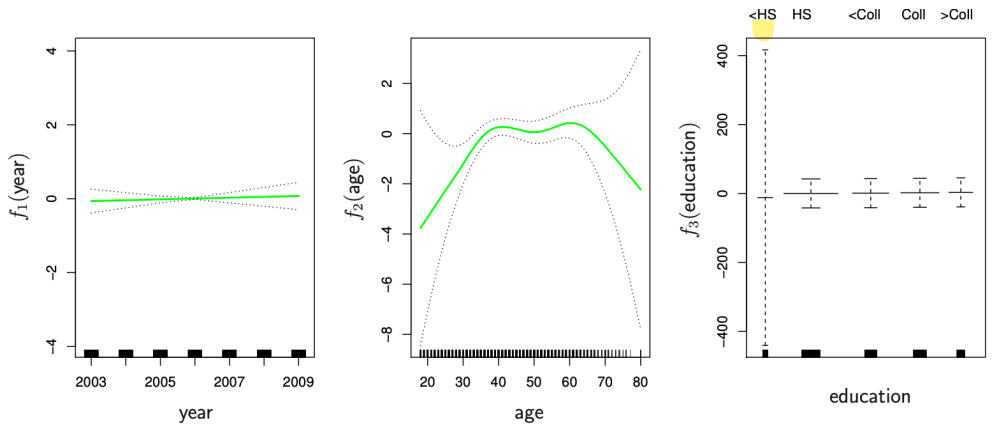
$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p)$$

GAM: Classification

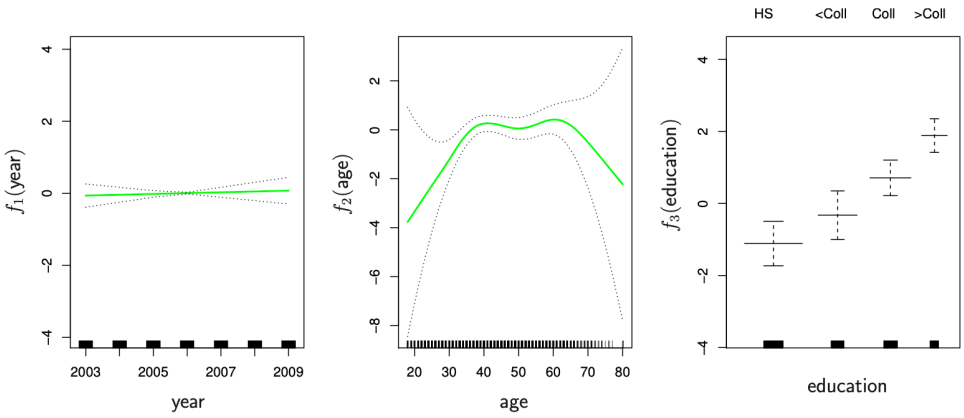
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Data issue!



GAM: Classification

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Pros and Cons of GAM

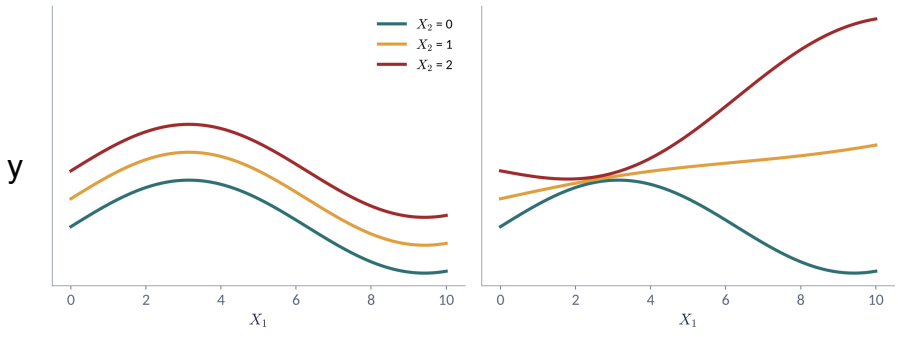
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- Fits a non-linear f(.) to each X_j automatically: so that we can automatically model non-linear relationships that standard linear regression will miss. This means that we do not need to manually try out many different transformations on each variable individually.
- The non-linear fits can potentially make more accurate predictions for the response Y.
- Because the model is additive, we can examine the effect of each X on Y individually while holding all of the other variables fixed.
- The smoothness of the function f(.) for the variable X_j can be summarized via degrees of freedom.

Pros and Cons of GAM

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- The main limitation of GAMs is that the model is restricted to be additive. With many variables, important interactions can be missed.



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year	Year that the wage information was recorded (2003–2009).
age	Age of the worker in years.
maritl	Marital status.
race	Race of the worker.
education	Highest level of education attained.
region	Region of the country.
jobclass	Type of job.
health	Health level of the worker.
health_ins	Does the worker have health insurance?
logwage	The natural logarithm of the worker's wage.
wage	The worker's raw wage in thousands of dollars.

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Lab: Nonlinear Models

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Lab Exercise:
Nonlinear Models

ISLR2::Auto

mpg	Fuel economy in miles per gallon
cylinders	Number of engine cylinders
displacement	Engine displacement in cubic inches
horsepower	Engine horsepower
weight	Vehicle weight in pounds
acceleration	Time to accelerate from 0 to 60 mph in seconds
year	Model year
origin	Origin of the car
name	Vehicle name and model

Lab Exercise:
Nonlinear Models

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- You can **choose any two nonlinear models** that you think is useful, including:
  - polynomial regression
  - step functions
  - regression splines
  - natural splines
  - smoothing splines
  - local regression
  - generalized additive models.
- Complete the lab check-in on Gradescope.
- **You can work in groups ($n \leq 3$)!**